STAT 331 Applied Linear Models

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Contents

1 Simple Linear Regression (SLR)

Lecture 1, 2025/09/04

Definition 1.1 (Linear Regression Model).

We say the regression model

$$y = f(x_1, x_2, \dots, x_p) + \epsilon$$

is **linear** if it is linear in the parameters, i.e. $\frac{\partial f}{\partial \beta_i}$ does not depend on β_i for $i = 0, 1, \dots, p$.

Lecture 2, 2025/09/09

1.1 Introduction

Definition 1.2 (SLR Population Model).

Population model.

$$\text{Response: } y = \beta_0 + \beta_1 x + \epsilon,$$

where β_0, β_1 are regression coefficients, x is the predictor, and ϵ is the random error.

Definition 1.3 (SLR Sample Model).

Sample: n pairs of observations (x_i, y_i), $i = 1, 2, \dots, n$.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where β_0 is the intercept and β_1 is the slope.

	Fixed/Random	Known/Unknown
x_i's	Fixed	Known
β_i's	Fixed	Unknown
ϵ_i's	Random	Unknown
y_i's	Random	Known

SLR Assumptions

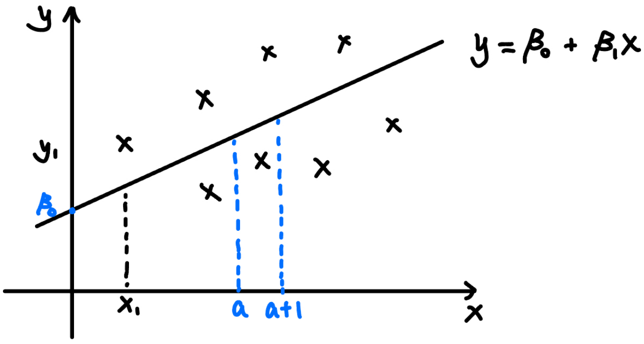
(1) $\mathbb{E}[\epsilon_i] = 0 \implies \mathbb{E}[y_i] = \beta_0 + \beta_1 x_i$.

(2) $\epsilon_1, \ldots, \epsilon_n$ are statistically independent.

(3) $\mathrm{Var}(\epsilon_i) = \sigma^2 \implies \mathrm{Var}(y_i) = \sigma^2$.

(4) ϵ_i is normally distributed, i.e. $\epsilon_i \sim N(0, \sigma^2) \implies y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$.

Note. The first three assumptions are called the **Gauss-Markov assumptions**.



The slope β_1 is often of primary interest. We have the following interpretations of β_1.

(1) $\beta_1 = \mathbb{E}[y \mid x = a+1] - \mathbb{E}[y \mid x = a]$.

(2) If $\beta_1 = 0$, then $\mathbb{E}[y \mid x] = \beta_0$, i.e. y does not depend on x.

1.2 Least Squares Estimation

Task: Given the sample observation (x_i, y_i), $i = 1, 2, \ldots, n$, we want to find (β_0, β_1) such that

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

is minimized.

To minimize the objective function $S(\beta_0, \beta_1)$, we set the partial derivatives to zero:

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = -2\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i) = 0, \tag{1.1}$$

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_1} = -2\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)x_i = 0. \tag{1.2}$$

Then, to solve for β_0, β_1,

(1.1) $\implies n\beta_0 = \sum_{i=1}^{n} y_i - \beta_1 \sum_{i=1}^{n} x_i$, so

$$\beta_0 = \bar{y} - \beta_1 \bar{x}, \text{ where } \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \text{ and } \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

(1.2) $\implies \sum_{i=1}^{n} x_i y_i - \beta_0 \sum_{i=1}^{n} x_i - \beta_1 \sum_{i=1}^{n} x_i^2 = 0$. Substitute β_0 into this, we have

$$\begin{aligned}
\sum_{i=1}^{n} x_i y_i - (\bar{y} - \beta_1 \bar{x})\sum_{i=1}^{n} x_i - \beta_1 \sum_{i=1}^{n} x_i^2 &= 0 \\
\sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i + \beta_1 \bar{x}\sum_{i=1}^{n} x_i - \beta_1 \sum_{i=1}^{n} x_i^2 &= 0 \\
\beta_1 = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} &= \frac{S_{xy}}{S_{xx}},
\end{aligned}$$

Lecture 3, 2025/09/11

Recall from STAT 231:

$$\text{Sample variance of } x\text{: } S_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2,$$

$$\text{Sample covariance: } q_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}).$$

Then,

$$\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{q_{xy}}{S_x^2}.$$

Definition 1.4 (**Pearson Correlation Coefficient**).

The **Pearson correlation coefficient** between x and y is

$$\rho_{xy} = \frac{\mathrm{Cov}\,(X, Y)}{\sqrt{\mathrm{Var}\,(X)\,\mathrm{Var}\,(Y)}}.$$

Definition 1.5 (**Sample Correlation Coefficient**).

The **sample correlation coefficient** between x and y is

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

Then, we have $\beta_1 = r_{xy}\frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}}$.

Thus, the least squares estimates (LSEs) are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}, \quad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Proposition 1.1 (**Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$**).

(1) The lease squares estimates (LSEs) are unbised, i.e. $\mathbb{E}\left[\hat{\beta}_0\right] = \beta_0, \mathbb{E}\left[\hat{\beta}_1\right] = \beta_1$.

(2) $\mathrm{Var}\left(\hat{\beta}_0\right) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right), \mathrm{Var}\left(\hat{\beta}_1\right) = \frac{\sigma^2}{S_{xx}}$.

(3) $\mathrm{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) = -\frac{\sigma^2\bar{x}}{S_{xx}}$.

Proof.

(1) We have

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}_1\right] = \mathbb{E}\left[\frac{S_{xy}}{S_{xx}}\right] &= \mathbb{E}\left[\frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{S_{xx}}\right] \\
&= \mathbb{E}\left[\frac{\sum_{i=1}^{n}(x_i - \bar{x})y_i}{S_{xx}}\right]
\end{aligned}$$

Note that $\hat{\beta}_1 = \sum_{i=1}^{n} c_i y_i$, where $c_i = \dfrac{x_i - \bar{x}}{S_{xx}}$.

Remark (**Properties of c_i's**).

(i) $\sum_{i=1}^n c_i = 0$.

(ii) $\sum_{i=1}^n c_i x_i = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) x_i = \frac{1}{S_{xx}} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right) = 1$.

(iii) $\sum_{i=1}^n c_i^2 = \frac{1}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{S_{xx}}$.

Then,

$$
\begin{aligned}
\mathbb{E}\left[\hat{\beta}_1\right] &= \mathbb{E}\left[\sum_{i=1}^{n} c_i y_i\right] = \sum_{i=1}^{n} c_i \mathbb{E}\left[y_i\right] \\
&= \sum_{i=1}^{n} c_i(\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^{n} c_i + \beta_1 \sum_{i=1}^{n} c_i x_i = \beta_1.
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
\mathbb{E}\left[\hat{\beta}_0\right] &= \mathbb{E}\left[\bar{y} - \hat{\beta}_1 \bar{x}\right] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[y_i\right] - \bar{x}\mathbb{E}\left[\hat{\beta}_1\right] \\
&= \frac{1}{n} \sum_{i=1}^{n} (\beta_0 + \beta_1 x_i) - \bar{x}\beta_1 \\
&= \beta_0 + \beta_1 \bar{x} - \bar{x}\beta_1 = \beta_0.
\end{aligned}
$$

(2) We have

$$
\begin{aligned}
\operatorname{Var}\left(\hat{\beta}_1\right) &= \operatorname{Var}\left(\sum_{i=1}^{n} c_i y_i\right) = \sum_{i=1}^{n} c_i^2 \operatorname{Var}\left(y_i\right) \\
&= \sigma^2 \sum_{i=1}^{n} c_i^2 = \frac{\sigma^2}{S_{xx}}.
\end{aligned}
$$

Similarly,

$$\begin{aligned}
\operatorname{Var}(\hat{\beta}_0) &= \operatorname{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) = \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} y_i - \bar{x}\sum_{i=1}^{n} c_i y_i\right) \\
&= \operatorname{Var}\left(\sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) y_i\right) = \sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right)^2 \operatorname{Var}(y_i) \\
&= \sigma^2 \sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right)^2 = \sigma^2\left(\sum_{i=1}^{n}\frac{1}{n^2} - 2\frac{\bar{x}}{n}\sum_{i=1}^{n} c_i + \bar{x}^2\sum_{i=1}^{n} c_i^2\right) \\
&= \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right).
\end{aligned}$$

(3) We have

$$\begin{aligned}
\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \operatorname{Cov}(\bar{y} - \hat{\beta}_1\bar{x}, \hat{\beta}_1) \\
&= \operatorname{Cov}\left(\sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) y_i, \sum_{i=1}^{n} c_i y_i\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) c_j \operatorname{Cov}(y_i, y_j)
\end{aligned}$$

Note that $\operatorname{Cov}(y_i, y_j) = \begin{cases} \sigma^2, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$ Then,

$$\begin{aligned}
\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \sigma^2 \sum_{i=1}^{n}\left(\frac{1}{n} - c_i\bar{x}\right) c_i \\
&= \sigma^2\left(\frac{1}{n}\sum_{i=1}^{n} c_i - \bar{x}\sum_{i=1}^{n} c_i^2\right) = -\frac{\sigma^2\bar{x}}{S_{xx}}.
\end{aligned}$$

□

Definition 1.6 (**Residual**).

The **residual** for the i-th observation is the difference between the observed and the fitted value:

$$r_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

Under the least squares fit, we have the following.

Proposition 1.2 (Properties of Residuals).

(1) $\sum_{i=1}^{n} r_i = 0.$

(2) $\sum_{i=1}^{n} r_i x_i = 0.$

(3) $\sum_{i=1}^{n} r_i \hat{y}_i = 0$, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

Lecture 4, 2025/09/16

To find the estimator of σ^2, note that

$$\begin{aligned} \epsilon_i &= y_i - (\beta_0 + \beta_1 x_i) \\ r_i &= y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i). \end{aligned}$$

If ϵ_i's are known, we can estimate σ^2 by $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} \epsilon_i^2$ because $\epsilon_i \sim N(0, \sigma^2)$ and

$$\mathbb{E}\left[\hat{\sigma}^2\right] = \mathbb{E}\left[\epsilon_i\right]^2 + \mathrm{Var}\left(\epsilon_i\right) = \sigma^2.$$

However $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} r_i^2\right] \neq \sigma^2$. If we define

$$s^2 = \frac{1}{n-2}\sum_{i=1}^{n} r_i^2,$$

then, $\mathbb{E}\left[s^2\right] = \sigma^2$.

Note. The degree of freedom is $n-2$ because we have estimated two parameters, β_0 and β_1.

Example (Degree of Freedom Example).
If $y_i \sim N(\mu, \sigma^2)$, then the sample variance estimator is

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \hat{\mu})^2,$$

where the degree of freedom is $n-1$ because we have estimated one parameter. Note that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$.

1.3 SLR: Inference

Proposition 1.3. Under the assumptions that ϵ_i's are independent and normally distributed,

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right).$$

Remark. If ϵ^2 is known, then

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0,1).$$

In most practice, σ^2 is unknown, and we replace σ^2 by s^2:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{s^2/S_{xx}}} \sim t_{n-2}.$$

Proposition 1.4 (Confidence Interval for β_1).
A $100(1-\alpha)\%$ confidence interval (CI) for β_1 is

$$\left[\hat{\beta}_1 - t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right), \hat{\beta}_1 + t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right)\right],$$

or equivalently

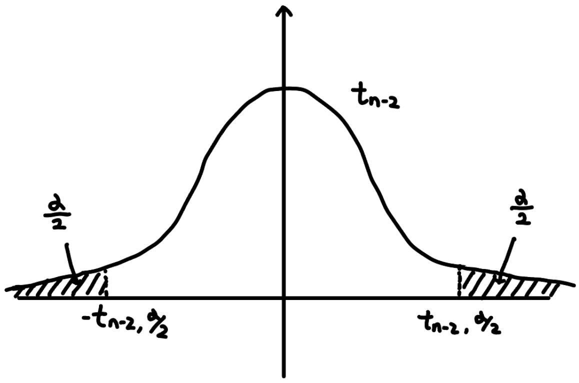
$$\hat{\beta}_1 \pm t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right), \text{ where } \mathrm{SE}\left(\hat{\beta}_1\right) = \frac{s}{\sqrt{S_{xx}}}.$$

Proof. Note that

$$\mathbb{P}\left(-t_{n-2,\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\mathrm{SE}\left(\hat{\beta}_1\right)} \leq t_{n-2,\alpha/2}\right) = 1-\alpha$$

$$\mathbb{P}\left(\hat{\beta}_1 - t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right) \leq \beta_1 \leq \hat{\beta}_1 + t_{n-2,\alpha/2}\,\mathrm{SE}\left(\hat{\beta}_1\right)\right) = 1-\alpha.$$

□



Hypothesis Testing for β_1

We test

$$H_0 : \beta_1 = \beta_1^* \text{ vs. } H_a : \beta_1 \neq \beta_1^*.$$

Under H_0, the test statistic is

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{\mathrm{SE}\left(\hat{\beta}_1\right)} \sim t_{n-2}.$$

Reject H_0 at significance level α if $|t| > t_{n-2,\alpha/2}$. Alternatively, compute the p-value:

$$p = 2\mathbb{P}(T > |t|) \quad \text{where } T \sim t_{n-2}$$

and reject H_0 if $p < \alpha$. Typically, we are interested in testing $H_0 : \beta_1 = 0$.

Note. $p = \mathbb{P}(|T| \geq |t|) = 2\mathbb{P}(T \geq |t|)$ because t-distribution is symmetric about 0.

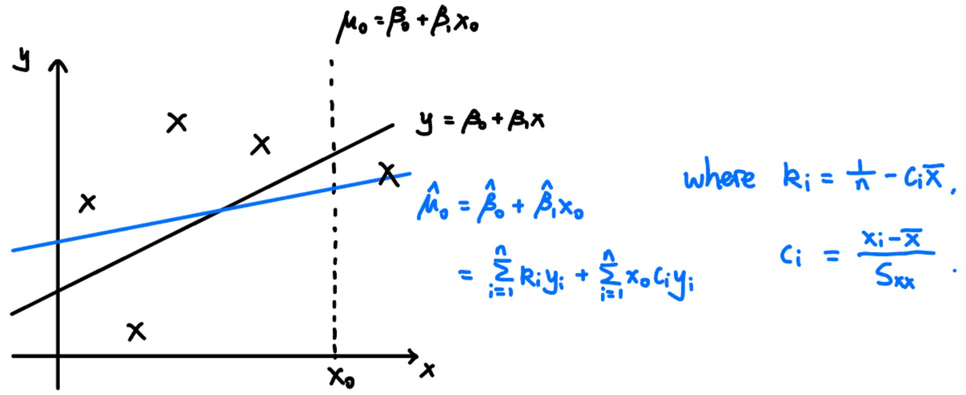
Inference for Mean Response

For the predictor value x_0, the mean response is $\mu_0 = \beta_0 + \beta_1 x_0$. We estimate μ_0 by

$$\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

We can use $\hat{\mu}_0 = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_0 = \bar{y} + \hat{\beta}_1(x_0 - \bar{x})$ to show that

$$\hat{\mu}_0 = \sum_i d_i y_i, \text{ where } d_i = \frac{1}{n} + \frac{(x_i - \bar{x})(x_0 - \bar{x})}{S_{xx}}.$$



Note. The black line is the true regression line and the blue line is the fitted line based on estimates.

Proposition 1.5 (**Expectation & Variance of** $\hat{\mu}_0$).

For $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$,

$$\mathbb{E}\left[\hat{\mu}_0\right] = \mu_0, \quad \operatorname{Var}\left(\hat{\mu}_0\right) = \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]\sigma^2.$$

Question: What is your best guess of y given $x = x_p$?

Answer: $\hat{y}_p$, where $\hat{y}_p = \hat{\beta}_0 + \hat{\beta}_1 x_p$.

Note that

$$\mathbb{E}\left[y_p - \hat{y}_p\right] = 0, \quad \operatorname{Var}\left(y_p - \hat{y}_p\right) = \left[1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right]\sigma^2.$$

Since σ^2 is unknown, we use s^2 and get

$$\frac{y_p - \hat{y}_p}{\operatorname{SE}\left(y_p - \hat{y}_p\right)} \sim t_{n-2}, \text{ where } \operatorname{SE}\left(y_p - \hat{y}_p\right) = s\sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}}.$$

Note. $y_p = \beta_0 + \beta_1 x_p + \epsilon_p$ is a new observation at $x = x_p$.

Proposition 1.6 (**Prediction Interval for** y_p).

A $100(1-\alpha)\%$ prediction interval (PI) for y_p is

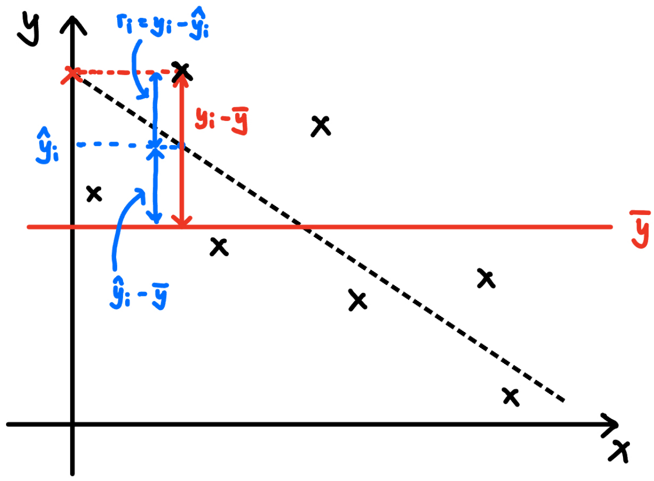
$$\hat{y}_p \pm t_{n-2,\alpha/2} \operatorname{SE}\left(y_p - \hat{y}_p\right), \text{ where } \operatorname{SE}\left(y_p - \hat{y}_p\right) = s\sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}}.$$

1.4 SLR: ANOVA and Cochran's Theorem

Analysis of Variance (ANOVA): How well does our regression model fit the response variable?

Total variation in y_i's is measured by the total sum of squares:

$$\mathrm{SST} = \sum_{i=1}^{n}(y_i - \bar{y})^2.$$



Proposition 1.7 (ANOVA Decomposition).
We have the following decomposition.

$$\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE},$$

where

$$\mathrm{SSR} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 \text{ is the regression sum of squares,}$$
$$\mathrm{SSE} = \sum_{i=1}^{n} r_i^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \text{ is the residual sum of squares.}$$

Proof. Notice that $y_i - \bar{y} = y_i - \hat{y}_i + \hat{y}_i - \bar{y}$. Now, we have

$$
\begin{aligned}
\text{SST} &= \sum_{i=1}^{n} (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + 2\sum_{i=1}^{n} (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\
&= \text{SSE} + \text{SSR} + 2\sum_{i=1}^{n} r_i (\hat{y}_i - \bar{y}) \\
&= \text{SSE} + \text{SSR}\,.
\end{aligned}
$$
□

Note. SSR is the variability explained by the model; SSE is the variability not explained by the model.

ANOVA for Testing $H_0 : \beta_1 = 0$

If H_0 is true, $y_i = \beta_0 + \epsilon_i$. SSR should be "small" relatively to SSE.

Proposition 1.8. Under $H_0 : \beta_1 = 0$, we have the following.

(1) $\frac{\text{SSR}}{\sigma^2} \sim \chi_1^2$.
(2) $\frac{\text{SST}}{\sigma^2} \sim \chi_{n-1}^2$.
(3) $\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-2}^2$, and SSE is independent of SSR.

Proof.

(1) We have

$$
\begin{aligned}
\text{SSR} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 &= \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y})^2 \\
&= \hat{\beta}_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{xx}.
\end{aligned}
$$

Recall that $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$. Then, we have $\left(\frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}}\right)^2 \sim \chi_1^2$. Under $H_0 : \beta_1 = 0$, we have

$$
\frac{\hat{\beta}_1^2 S_{xx}}{\sigma^2} \sim \chi_1^2 \implies \frac{\text{SSR}}{\sigma^2} \sim \chi_1^2.
$$

(2) Under $H_0 : \beta_1 = 0$, we have $y_1, \dots, y_n \overset{\text{iid}}{\sim} N(\beta_0, \sigma^2)$. That is, $\frac{y_i - \beta_0}{\sigma} \sim N(0,1)$. Thus,

$$\sum_{i=1}^{n} \left(\frac{y_i - \beta_0}{\sigma} \right)^2 \sim \chi_n^2.$$

Furthermore, $\left(\frac{\bar{y} - \beta_0}{\sigma/\sqrt{n}} \right)^2 \sim \chi_1^2$ because $\bar{y} \sim N\left(\beta_0, \frac{\sigma^2}{n}\right)$. Now,

$$\begin{aligned}
\text{SST} &= \sum_{i=1}^{n} (y_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} \left[(y_i - \beta_0) - (\bar{y} - \beta_0) \right]^2 \\
&= \sum_{i=1}^{n} (y_i - \beta_0)^2 - n(\bar{y} - \beta_0)^2.
\end{aligned}$$

Thus, we have

$$\frac{\sum_{i=1}^{n} (y_i - \beta_0)^2}{\sigma^2} = \frac{\text{SST}}{\sigma^2} + \left(\frac{\bar{y} - \beta_0}{\sigma/\sqrt{n}} \right)^2 \iff S = Q_1 + Q_2,$$

where $S \sim \chi_n^2$. For $Q_1 = \frac{\text{SST}}{\sigma^2}$, there are only $n-1$ degrees of freedom because $\sum_{i=1}^{n} (y_i - \bar{y}) = 0$. For $Q_2 = \left(\frac{\bar{y} - \beta_0}{\sigma/\sqrt{n}} \right)^2$, we only have $\bar{y}$ and so $d_2 = 1$. By Cochran's theorem, Q_1 and Q_2 are independent, and $\frac{\text{SST}}{\sigma^2} \sim \chi_{n-1}^2$.

(3) We have

$$\begin{aligned}
\frac{\text{SST}}{\sigma^2} &= \frac{\text{SSE}}{\sigma^2} + \frac{\text{SSR}}{\sigma^2} \\
\underbrace{S}_{\sim \chi_{n-1}^2} &= Q_1 + \underbrace{Q_2}_{\sim \chi_1^2}.
\end{aligned}$$

The degrees of freedom of $\text{SSE} = \sum_{i=1}^{n} r_i^2$ is $n-2$ because $\sum_{i=1}^{n} r_i = 0$ and $\sum_{i=1}^{n} r_i x_i = 0$. By Cochran's theorem, $\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-2}^2$ and SSE is independent of SSR.

□

> **Theorem 1.9 (Cochran's Theorem).**
> Suppose that $U_i \overset{\text{iid}}{\sim} N(0,1)$ for $i = 1, 2, \ldots, n$, and
>
> $$\sum_{i=1}^{n} U_i^2 = Q_1 + Q_2.$$
>
> Let d_1 and d_2 be the degrees of freedom of Q_1 and Q_2, i.e. the numbers of linearly independent linear combinations of the U_i's in Q_1 and Q_2. If $d_1 + d_2 = n$, then Q_1 and Q_2 are independent, and furthermore,
>
> $$Q_1 \sim \chi^2_{d_1}, \quad Q_2 \sim \chi^2_{d_2}.$$

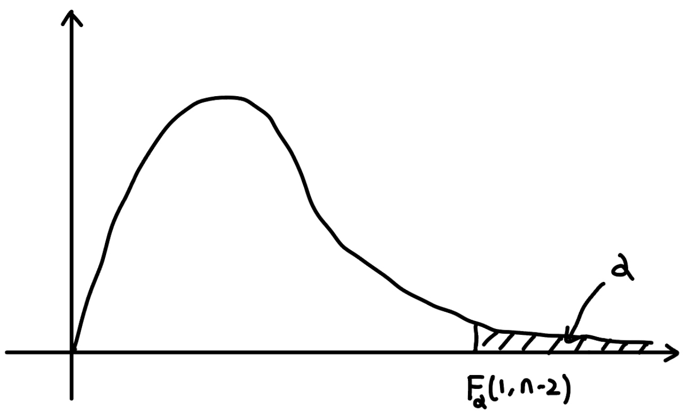
Proposition 1.8 above allows us to define the following F-statistic (with $\beta_1 = 0$ under H_0):

$$F = \frac{\frac{\text{SSR}}{\sigma^2}/1}{\frac{\text{SSE}}{\sigma^2}/(n-2)} = \frac{\text{SSR}}{\text{SSE}/(n-2)} = \frac{\text{MSR}}{\text{MSE}} \sim F(1, n-2).$$

Note that MSR is the mean squares of regression and MSE is the mean squares of error. To test $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$, we reject H_0 at the level α if

$$F > F_\alpha(1, n-2), \text{ where } F_\alpha(1, n-2) \text{ is the } 100(1-\alpha)\%\text{-tile of } F(1, n-2).$$

The F-distribution looks like this:



To test $H_0 : \beta_1 = 0$, we use the F-statistic:

$$F = \frac{\text{SSR}}{\text{SSE}\,/(n-2)} = \frac{\text{MSR}}{\text{MSE}} \sim F(1, n-2).$$

Recall that

$$\frac{\hat{\beta}_1 - \beta_1^*}{\text{SE}\,(\hat{\beta}_1)} \sim t_{n-2}$$

and we can use this to test $H_0 : \beta_1 = \beta_1^*$. In SLR, testing $H_0 : \beta_1 = 0$ using t-test and F-test are equivalent. If $X \sim t_{n-2}$, then $X^2 \sim F(1, n-2)$.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F
Regression	SSR	1	$\text{MSR} = \frac{\text{SSR}}{1}$	$\frac{\text{MSR}}{\text{MSE}}$
Residual	SSE	$n-2$	$\text{MSE} = \frac{\text{SSE}}{n-2}$	
Total	SST	$n-1$		

Definition 1.7 (**Coefficient of Determination**).
The **coefficient of determination** is defined as

$$R^2 = \frac{\text{SSR}}{\text{SST}}.$$

It measures the proportion of total variation in the response variable y around the mean that is explained by the regression model.

$0 \le R^2 \le 1$, $R^2 = 0$ means none of the variability in y is explained by x. In SLR, we have

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\hat{\beta}_1^2 S_{xx}}{S_{yy}} = \frac{\left(\frac{S_{xy}}{S_{xx}}\right)^2 S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{yy} S_{xx}}.$$

Recall that the sample correlation coefficient is

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

and thus $R^2 = r^2$.

2 Multiple Linear Regression (MLR)

2.1 Linear Algebra Review and MVN Distribution

Definition 2.1 (Random Vectors).

Let $\mathbf{Y} = (y_1, \ldots, y_n)^\top$ be a vector of random variables. Then, $\mathbf{Y}$ is called a **random vector**. Note that

$$\mathbb{E}[y_i] = \mu_i, \quad \operatorname{Var}(y_i) = \sigma_i^2, \quad \operatorname{Cov}(y_i, y_j) = \sigma_{ij}.$$

Also,

$$\mathbb{E}[\mathbf{Y}] = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}, \quad \operatorname{Var}(\mathbf{Y}) = \operatorname{Cov}(\mathbf{Y}, \mathbf{Y}) = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{bmatrix}.$$

Remark. Note that

$$\begin{aligned} \operatorname{Var}(\mathbf{Y}) &= \mathbb{E}\left[(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])^\top\right] \\ &= \begin{bmatrix} \operatorname{Var}(y_1) & \operatorname{Cov}(y_1, y_2) & \cdots & \operatorname{Cov}(y_1, y_n) \\ \operatorname{Cov}(y_2, y_1) & \operatorname{Var}(y_2) & \cdots & \operatorname{Cov}(y_2, y_n) \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{Cov}(y_n, y_1) & \operatorname{Cov}(y_n, y_2) & \cdots & \operatorname{Var}(y_n) \end{bmatrix} \end{aligned}$$

Example. If $y_1, \ldots, y_n$ are independent, $\operatorname{Cov}(y_i, y_j) = 0 \ \forall i \neq j$. Furthermore, $\operatorname{Var}(y_i) = \sigma^2$ $\forall i$. This implies that

$$\operatorname{Var}(\mathbf{Y}) = \sigma^2 I, \qquad \text{where } I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \text{ is the identity matrix.}$$

Proposition 2.1 (**Basic Properties**).

Suppose $A = (a_{ij})_{m\times n}$, $\mathbf{b} = (b_1, \dots, b_m)^\top$, $\mathbf{c} = (c_1, \dots, c_m)^\top$ are matrix and vectors of constants. Then,

(1) $\mathbb{E}[A\mathbf{Y} + \mathbf{b}] = A\mathbb{E}[\mathbf{Y}] + \mathbf{b}$.

(2) $\mathrm{Var}(\mathbf{Y} + \mathbf{c}) = \mathrm{Var}(\mathbf{Y})$.

(3) $\mathrm{Var}(A\mathbf{Y}) = A\,\mathrm{Var}(\mathbf{Y})\,A^\top$.

(4) $\mathrm{Var}(A\mathbf{Y} + \mathbf{b}) = A\,\mathrm{Var}(\mathbf{Y})\,A^\top$.

Definition 2.2 (**Differentiation: Linear & Quadratic Forms**).

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a function and $\mathbf{Y}$ be a random vector. Then,

(1) If $f = f(\mathbf{Y})$, then

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{Y}} = \begin{bmatrix} \frac{\mathrm{d}f}{\mathrm{d}y_1} \\ \vdots \\ \frac{\mathrm{d}f}{\mathrm{d}y_n} \end{bmatrix}.$$

(2) If $f = \mathbf{c}^\top\mathbf{Y} = \sum c_i y_i$, then

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{Y}} = \mathbf{c}.$$

(3) If $A = A^\top$ and $f = \mathbf{Y}^\top A\mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} y_i y_j$, then

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{Y}} = 2A\mathbf{Y}.$$

In this case, f is called a **quadratic form**.

Definition 2.3 (**Trace**).

The **trace** of a square matrix $A = (a_{ij})_{n\times n}$ is defined as

$$\mathrm{tr}(A) = \sum_{i=1}^n a_{ii}.$$

Remark. We have $\mathrm{tr}(BC) = \mathrm{tr}(CB)$.

Definition 2.4 (Rank).

The **rank** of a matrix A is defined as

$$\text{rank}(A) = \text{the maximum number of linearly independent columns (or rows) of } A.$$

Definition 2.5 (Linearly Independence).

Random vectors $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are linearly independent $\iff$

$$c_1\mathbf{Y}_1 + \cdots + c_n\mathbf{Y}_n = 0 \implies c_1 = c_2 = \cdots = c_n = 0.$$

Definition 2.6 (Orthogonality, Orthogonal Matrix).

Random vectors $\mathbf{X}, \mathbf{Y}$ are **orthogonal** if $\mathbf{X}^\top \mathbf{Y} = 0$.

A square matrix A is an **orthogonal matrix** if $A^\top A = AA^\top = I$.

Definition 2.7 (Eigenvalues, Eigenvectors).

Let A be a square matrix. A scalar λ is an **eigenvalue** of A if $\exists$ a non-zero vector $\mathbf{v}$ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The vector $\mathbf{v}$ is called an **eigenvector** corresponding to the eigenvalue λ.

Definition 2.8 (Idempotent Matrix).

A square matrix A is **idempotent** if $A^2 = A$.

Proposition 2.2. Let A be an idempotent matrix.

(1) All eigenvalues of A are either 0 or 1.

(2) If A is symmetric (i.e. $A = A^\top$), then $\exists$ orthogonal P s.t.

$$A = P\Lambda P^\top, \quad \text{where } \Lambda = \text{diag}(1, \dots, 1, 0, \dots, 0).$$

Then, $\text{tr}(A) = \text{rank}(A) = \#$ of eigenvalues equal to 1.

Proof.

(1) Let λ be an eigenvalue of A with corresponding eigenvector $\mathbf{v}$. Then,

$$A^2\mathbf{v} = A(A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda A\mathbf{v} = \lambda^2\mathbf{v}.$$

But since A is idempotent, $A^2\mathbf{v} = A\mathbf{v} = \lambda\mathbf{v}$. Thus,

$$\lambda^2\mathbf{v} = \lambda\mathbf{v} \implies \lambda(\lambda - 1)\mathbf{v} = 0.$$

(2) For the $\text{tr}(A) = \text{rank}(A)$ part, note that

$$\text{tr}(A) = \text{tr}\left(P\Lambda P^\top\right) = \text{tr}\left(\Lambda P^\top P\right) = \text{tr}(\Lambda) = \text{rank}(A).$$

□

Definition 2.9 (**Multivariate Normal Distribution (MVN)**).

$\mathbf{Y} \sim \text{MVN}(\boldsymbol{\mu}, \Sigma)$ if $\mathbf{Y}$ has the following probability density function (PDF):

$$f(\mathbf{y}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu})\right).$$

Also, $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$ and $\text{Var}(\mathbf{Y}) = \Sigma$.

Proposition 2.3 (**Properties of MVN**).

Suppose $\mathbf{Y}$ is a random vector and A, B are matrices. Then,

(1) $y_1, \dots, y_n$ are independent $\iff$ Σ is diagonal.

(2) Marginal normality: $y_i \sim \text{N}(\mu_i, \sigma_{ii})$.

(3) If $\mathbf{Y} \sim \text{MVN}(\boldsymbol{\mu}, \Sigma)$ and $Z = A\mathbf{Y}$, then

$$Z \sim \text{MVN}(A\boldsymbol{\mu}, A\Sigma A^\top).$$

(4) If $\mathbf{Y} \sim \text{MVN}(0, \sigma^2 I)$, then

$$\frac{1}{\sigma^2}\mathbf{Y}^\top\mathbf{Y} \sim \chi^2_n.$$

(5) If $U = A\mathbf{Y}$ and $W = B\mathbf{Y}$, then

$$\mathrm{Cov}\,(U, W) = A\Sigma B^\top \quad \text{i.e. } \mathrm{Cov}\,(A\mathbf{Y}, B\mathbf{Y}) = A\,\mathrm{Var}\,(\mathbf{Y})\,B^\top.$$

Then, $U \perp\!\!\!\perp W \iff A\Sigma B^\top = 0$.

Remark. For (5), note that

$$\mathrm{Cov}\,(U, W) = \mathbb{E}\left[(U - \mathbb{E}\,[U])(W - \mathbb{E}\,[W])^\top\right] = A\mathbb{E}\left[(\mathbf{Y} - \mathbb{E}\,[\mathbf{Y}])(\mathbf{Y} - \mathbb{E}\,[\mathbf{Y}])^\top\right]B^\top = A\Sigma B^\top.$$

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2.2 Introduction to MLR

Definition 2.10 (Multiple Linear Regression (MLR)).
The **MLR** model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n,$$

Explanation of the regression coefficients $\beta_1, \dots, \beta_p$:

- β_j: the average amount of increase (decrease) in response when the ith predictor x_j increases (decreases) by 1 unit while holding all other predictors fixed/constant.

The null hypothesis $H_0 : \beta_j = 0$ means that x_j is not (linearly) related to y given all other predictors in the model.

Matrix Form Representation

We can write the n simultaneous linear models as

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}_{n\times(p+1)} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}_{(p+1)\times 1} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}_{n\times 1}$$

Equivalently, we can write

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

And

$$\boldsymbol{\epsilon} \sim \text{MVN}(0, \sigma^2 I), \quad \mathbf{Y} \sim \text{MVN}(X\boldsymbol{\beta}, \sigma^2 I).$$

2.3 Least Squares Estimation (LSE)

Least Squares Estimation (LSE) of $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^\top$

Once we collect data and observe y_i's, we minimize

$$\begin{aligned} S(\boldsymbol{\beta}) &= \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_p x_{ip})^2 \\ &= (\mathbf{Y} - X\boldsymbol{\beta})^\top (\mathbf{Y} - X\boldsymbol{\beta}) \\ &= \mathbf{Y}^\top \mathbf{Y} - \underbrace{\mathbf{Y}^\top}_{1\times n} X\boldsymbol{\beta} - \boldsymbol{\beta}^\top X^\top \mathbf{Y} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}. \end{aligned}$$

Note that

$$\mathbf{Y}^\top X\boldsymbol{\beta} = (\mathbf{Y}^\top X\boldsymbol{\beta})^\top = \boldsymbol{\beta}^\top X^\top \mathbf{Y} \quad \text{is a scalar.}$$

Thus,

$$S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\underbrace{\boldsymbol{\beta}^\top X^\top \mathbf{Y}}_{\text{linear}} + \underbrace{\boldsymbol{\beta}^\top X^\top X\beta}_{\text{quadratic form}}.$$

Recall that

$$\frac{\mathrm{d}\boldsymbol{\beta}^\top \mathbf{Y}}{\mathrm{d}\mathbf{Y}} = \boldsymbol{\beta}, \quad \frac{\mathrm{d}\mathbf{Y}^\top A\mathbf{Y}}{\mathrm{d}\mathbf{Y}} = 2A\mathbf{Y}.$$

Taking derivative w.r.t. $\boldsymbol{\beta}$, we have

$$\frac{\mathrm{d}S(\boldsymbol{\beta})}{\mathrm{d}\boldsymbol{\beta}} = -2X^\top \mathbf{Y} + 2X^\top X\boldsymbol{\beta} \overset{\text{set}}{=} 0 \implies \hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{Y}.$$

Note. We assume that $(X^\top X)$ is invertible, i.e. $\operatorname{rank}(X) = p + 1$.

Proposition 2.4 (**Properties of LSE $\hat{\beta}$**).

(1) $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$, i.e. $\mathbb{E}\left[\hat{\boldsymbol{\beta}}\right] = \boldsymbol{\beta}$.

(2) $\operatorname{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2 (X^\top X)^{-1}$.

Proof.

(1) We have

$$\mathbb{E}\left[\hat{\boldsymbol{\beta}}\right] = \mathbb{E}\left[(X^\top X)^{-1}X^\top \mathbf{Y}\right] = (X^\top X)^{-1}X^\top \mathbb{E}\left[\mathbf{Y}\right] = (X^\top X)^{-1}X^\top (X\boldsymbol{\beta}) = \boldsymbol{\beta}.$$

(2) We have

$$\begin{aligned} \operatorname{Var}\left(\hat{\boldsymbol{\beta}}\right) &= \operatorname{Var}\left((X^\top X)^{-1}X^\top \mathbf{Y}\right) \\ &= (X^\top X)^{-1}X^\top \operatorname{Var}\left(\mathbf{Y}\right) X \left[(X^\top X)^{-1}\right]^\top \qquad \text{since } \operatorname{Var}\left(A\mathbf{Y}\right) = A \operatorname{Var}\left(\mathbf{Y}\right) A^\top \end{aligned}$$

Note that $\left[(X^\top X)^{-1}\right]^\top = \left[(X^\top X)^\top\right]^{-1} = (X^\top X)^{-1}$, see remark below.

$$\begin{aligned} &= (X^\top X)^{-1}X^\top (\sigma^2 I) X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1}X^\top X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1}. \end{aligned}$$

Remark. In general, $A^{-1}A = I \implies (A^{-1}A)^\top = A^\top (A^{-1})^\top = I^\top = I \implies (A^{-1})^\top = (A^\top)^{-1}$.

□

Proposition 2.5.

(1) Fitted values:

$$\hat{\mathbf{Y}} = X\hat{\boldsymbol{\beta}} = \underbrace{X(X^\top X)^{-1}X^\top}_{H} \mathbf{Y} = H\mathbf{Y}, \quad \text{where } H \text{ is the hat/projection matrix.}$$

(2) H is idempotent and symmetric.

Proof.

(1) Simply substitute the LSE $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \mathbf{Y}$.

(2) Note that

$$HH = X(X^\top X)^{-1} \underbrace{X^\top X (X^\top X)^{-1}}_{I} X^\top = X(X^\top X)^{-1}X^\top = H.$$

□

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Definition 2.11 (**Residuals**).

The **residuals** in MLR are defined as

$$\mathbf{r} = \mathbf{Y} - \hat{\mathbf{Y}} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}.$$

Note that $\mathbf{r} = \mathbf{Y} - H\mathbf{Y} = (I - H)\mathbf{Y}$ where $I - H$ is also idempotent and symmetric.

Proposition 2.6 (**Properties of Residuals**).

(1) $\sum_{i=1}^{n} r_i = 0$.

(2) $X^\top \mathbf{r} = \vec{0}$.

(3) $\hat{\mathbf{Y}}^\top \mathbf{r} = 0$.

(4) $\mathbb{E}[\mathbf{r}] = \vec{0}$.

(5) $\operatorname{Var}(\mathbf{r}) = \sigma^2(I - H)$.

Proof.

(1) Since the first row of $X^\top$ consists of all 1's, so $X^\top \mathbf{r} = \vec{0}$ implies $\sum r_i = 0$.

(2) Note that

$$\begin{aligned} X^\top \mathbf{r} &= X^\top(\mathbf{Y} - \hat{\mathbf{Y}}) \\ &= X^\top \mathbf{Y} - X^\top H \mathbf{Y} \\ &= X^\top \mathbf{Y} - \underbrace{X^\top X (X^\top X)^{-1}}_{I} X^\top \mathbf{Y} \\ &= X^\top \mathbf{Y} - X^\top \mathbf{Y} = \vec{0}_{(p+1)\times 1}. \end{aligned}$$

(3) Note that

$$\hat{\mathbf{Y}}^\top \mathbf{r} = (X\hat{\boldsymbol{\beta}})^\top \mathbf{r} = \hat{\boldsymbol{\beta}}^\top X^\top \mathbf{r} = \hat{\boldsymbol{\beta}}^\top \vec{0} = 0.$$

(4) Note that

$$\begin{aligned}
\mathbb{E}\left[\mathbf{r}\right] &= \mathbb{E}\left[(I-H)\mathbf{Y}\right] \\
&= (I-H)\mathbb{E}\left[\mathbf{Y}\right] \\
&= (I - X(X^\top X)^{-1}X^\top)X\boldsymbol{\beta} \\
&= X\boldsymbol{\beta} - X\underbrace{(X^\top X)^{-1}X^\top X}_{I}\boldsymbol{\beta} = \vec{0}_{n\times 1}.
\end{aligned}$$

(5) Note that

$$\begin{aligned}
\operatorname{Var}\left(\mathbf{r}\right) &= \operatorname{Var}\left((I-H)\mathbf{Y}\right) \\
&= (I-H)\operatorname{Var}\left(\mathbf{Y}\right)(I-H)^\top \\
&= (I-H)(\sigma^2 I)(I-H) \qquad \text{since } I-H \text{ is symmetric} \\
&= \sigma^2(I-H)(I-H) \qquad \text{since } I-H \text{ is idempotent} \\
&= \sigma^2(I-H).
\end{aligned}$$

□

An Estimator of σ^2

Proposition 2.7. We have that

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n r_i^2}{n-(p+1)}$$

is an unbiased estimator of σ^2, i.e. $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$.

Note.

(1) In SLR, $p=1$, we had $\hat{\sigma}^2 = \frac{\sum r_i^2}{n-2}$ where $n-2$ is the degrees of freedom.

(2) Why $n-(p+1)$?

$p+1$ is the # of parameters need to be estimated.

We lose $p+1$ degrees of freedom because we estimate $p+1$ parameters $\beta_0, \dots, \beta_p$. The constraints are $X^\top \mathbf{r} = \vec{0}$ which gives $\sum r_i = 0$ and $\sum x_{ji} r_i = 0$ for $j = 1, \dots, p$.

Proof. We have

$$
\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{n} r_i^2\right] &= \mathbb{E}\left[\mathbf{r}^\top \mathbf{r}\right] = \mathbb{E}\left[\operatorname{tr}\!\left(\mathbf{r}^\top \mathbf{r}\right)\right] \\
&= \mathbb{E}\left[\operatorname{tr}\!\left(\mathbf{r}\mathbf{r}^\top\right)\right] && \text{since } \operatorname{tr}(A) = \operatorname{tr}\!\left(A^\top\right) \text{ for any square matrix } A \\
&= \operatorname{tr}(\operatorname{Var}(\mathbf{r})) && \text{since } \mathbb{E}[\mathbf{r}] = \vec{0} \\
&= \operatorname{tr}\!\left(\sigma^2(I-H)\right) \\
&= \sigma^2 \left[\operatorname{tr}(I_{n\times n}) - \operatorname{tr}(H)\right] \\
&= \sigma^2 \left[n - \operatorname{tr}\!\left(X(X^\top X)^{-1}X^\top\right)\right] \\
&= \sigma^2 \left[n - \operatorname{tr}\!\left((X^\top X)^{-1}X^\top X\right)\right] && \text{since } \operatorname{tr}(BC) = \operatorname{tr}(CB) \\
&= \sigma^2 \left[n - \operatorname{tr}\!\left(I_{(p+1)\times(p+1)}\right)\right] \\
&= \sigma^2 \left[n - (p+1)\right].
\end{aligned}
$$

□

2.4 MLR: Inference

Sampling Distribution of $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ under normality assumption $\mathbf{Y} \sim \text{MVN}(X\boldsymbol{\beta}, \sigma^2 I)$.

Proposition 2.8.

(1) $\hat{\boldsymbol{\beta}} \sim \text{MVN}(\boldsymbol{\beta}, \sigma^2(X^\top X)^{-1})$.

(2) $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are independent.

(3) $\frac{(n-(p+1))\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p-1}$.

Proof.

(1) Recall that $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \mathbf{Y}$. We found the expectation and variance. It follows MVN because $\mathbf{Y} \sim \text{MVN}(X\boldsymbol{\beta}, \sigma^2 I)$ and $\hat{\boldsymbol{\beta}}$ is a linear transformation of $\mathbf{Y}$.

(2) Note that $\hat{\boldsymbol{\beta}} \sim \text{MVN}(\boldsymbol{\beta}, \sigma^2(X^\top X)^{-1})$ from (1) and $\mathbf{r} = (I-H)\mathbf{Y} \sim \text{MVN}(\vec{0}, \sigma^2(I-H))$. Then,

$$\begin{aligned}
\text{Cov}\left(\hat{\boldsymbol{\beta}}, \mathbf{r}\right) &= \text{Cov}\left((X^\top X)^{-1}X^\top \mathbf{Y}, (I-H)\mathbf{Y}\right) \\
&= (X^\top X)^{-1}X^\top \,\text{Var}\,(\mathbf{Y})\,(I-H)^\top \\
&= (X^\top X)^{-1}X^\top(\sigma^2 I)(I - X(X^\top X)^{-1}X^\top) \\
&= \sigma^2\left[(X^\top X)^{-1}X^\top - (X^\top X)^{-1}\underbrace{X^\top X(X^\top X)^{-1}}_{I}X^\top\right] \\
&= \sigma^2\left[(X^\top X)^{-1}X^\top - (X^\top X)^{-1}X^\top\right] \\
&= \mathcal{O}_{(p+1)\times n}.
\end{aligned}$$

Since $\hat{\boldsymbol{\beta}}$ and $\mathbf{r}$ are jointly normal, $\text{Cov}\left(\hat{\boldsymbol{\beta}}, \mathbf{r}\right) = 0$ implies that $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathbf{r}$. Since $\hat{\sigma}^2$ is a function of $\mathbf{r}$, we have $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \hat{\sigma}^2$.

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(3) Note that

$$\begin{aligned}
\frac{(n-(p+1))\hat{\sigma}^2}{\sigma^2} &= \frac{\mathbf{r}^\top \mathbf{r}}{\sigma^2} \\
&= \left(\frac{\mathbf{r}}{\sigma}\right)^\top \left(\frac{\mathbf{r}}{\sigma}\right) \\
&= {\mathbf{r}^*}^\top \mathbf{r}^* \qquad \text{where } \mathbf{r}^* = \frac{\mathbf{r}}{\sigma} = \frac{(I-H)\mathbf{Y}}{\sigma} \sim \text{MVN}(\vec{0}, I-H)
\end{aligned}$$

By the Eigen Decomposition of a symmetric and idempotent matrix, $\exists$ orthogonal P such that $I - H = P\Lambda P^\top$ where $\Lambda = \text{diag}(1, \dots, 1, 0, \dots, 0)$ with # of 1's $= \text{tr}(I-H) = n-(p+1)$. Now define a new random vector $\mathbf{z} = P^\top \mathbf{r}^*$. Then,
$\mathbf{z} \sim \text{MVN}(\vec{0}, P^\top P\Lambda P^\top P) = \text{MVN}(\vec{0}, \Lambda) \implies z_1, \dots, z_{n-p-1} \overset{iid}{\sim} \text{N}(0,1)$ and $z_{n-p}, \dots, z_n = 0$. Thus,

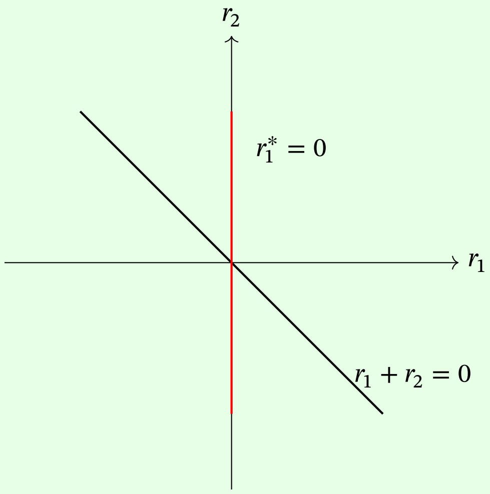
$$\frac{(n-(p+1))\hat{\sigma}^2}{\sigma^2} = (P\mathbf{z})^\top(P\mathbf{z}) = \mathbf{z}^\top P^\top P\mathbf{z} = \mathbf{z}^\top\mathbf{z} = \sum_{i=1}^{n-p-1} z_i^2 \sim \chi^2_{n-p-1}.$$

Note. Since P is orthogonal, $P^\top P = I$.

□

Example. Let $\mathbf{r} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$ with $r_1 + r_2 = 0$. Let

$$r^* = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \mathbf{r} = \frac{1}{\sqrt{2}} \begin{bmatrix} r_1 + r_2 \\ r_1 - r_2 \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} r_1 \end{bmatrix}.$$



This example shows some intuition for the proof of Proposition 2.8 (3).

Inference of a single β_i

From Proposition 2.8 (1), we have

$$\hat{\beta}_i \sim \mathrm{N}\left(\beta_i, \sigma^2 v_{ii}\right)$$

where v_{ii} is the ith diagonal element of $(X^\top X)^{-1}$.

From Proposition 2.8 (2) and (3), we have

$$T = \frac{\frac{\hat{\beta}_i - \beta_i}{\sqrt{\hat{\sigma}^2 v_{ii}}}}{\sqrt{\frac{(n-(p+1))\hat{\sigma}^2/\sigma^2}{n-(p+1)}}} \sim t_{n-p-1}$$

since $\frac{\hat{\beta}_i - \beta_i}{\sqrt{\hat{\sigma}^2 v_{ii}}} \sim \mathrm{N}(0,1)$ and $(n-(p+1))\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p-1}$.

We can further simplify T as

$$T = \frac{\hat{\beta}_i - \beta_i}{\sqrt{\hat{\sigma}^2 \nu_{ii}}} \sim t_{n-p-1}$$

and $\mathrm{SE}\left(\hat{\beta}_i\right) = \sqrt{\hat{\sigma}^2 \nu_{ii}}$.

Inference of a linear combination of β_i's

Let $\mathbf{a} = (a_0, a_1, \dots, a_p)^\top$ be a $(p+1)$-dimentional vector of constants. We are interested in the inference of

$$\theta = \mathbf{a}^\top \boldsymbol{\beta} = \sum_{i=0}^{p} a_i \beta_i.$$

We estimate θ by $\hat{\theta} = \mathbf{a}^\top \hat{\boldsymbol{\beta}}$.

Example. Let $p = 2$ and let $\mathbf{a}^\top = (1, 1, 5)$. Then,

$$\theta = \mathbf{a}^\top \boldsymbol{\beta} = \beta_0 + \beta_1 + 5\beta_2.$$

Recall that $\hat{\boldsymbol{\beta}} \sim \mathrm{MVN}(\boldsymbol{\beta}, \sigma^2 (X^\top X)^{-1})$. Then,

$$\hat{\theta} = \mathbf{a}^\top \hat{\boldsymbol{\beta}} \sim \mathrm{N}(\mathbf{a}^\top \boldsymbol{\beta}, \sigma^2 \mathbf{a}^\top (X^\top X)^{-1} \mathbf{a}).$$

We have,

$$\frac{\hat{\theta} - \theta}{\sigma \sqrt{\mathbf{a}^\top (X^\top X)^{-1} \mathbf{a}}} \sim \mathrm{N}(0, 1), \quad \text{and} \quad \frac{\hat{\theta} - \theta}{\hat{\sigma} \sqrt{\mathbf{a}^\top (X^\top X)^{-1} \mathbf{a}}} \sim t_{n-p-1}.$$

Prediction Interval

Task: Predict y for given values of predictor variables $x_1, \dots, x_p$.

Let $\mathbf{a}_p = (1, x_1, \dots, x_p)^\top$. Since $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$, we have

$$y_p = \mathbf{a}_p^\top \boldsymbol{\beta} + \epsilon_p.$$

We plug in the LSE $\hat{\boldsymbol{\beta}}$ to estimate y_p:

$$\hat{y}_p = \mathbf{a}_p^\top \hat{\boldsymbol{\beta}}.$$

Note that the prediction error is $y_p - \hat{y}_p$. Now,

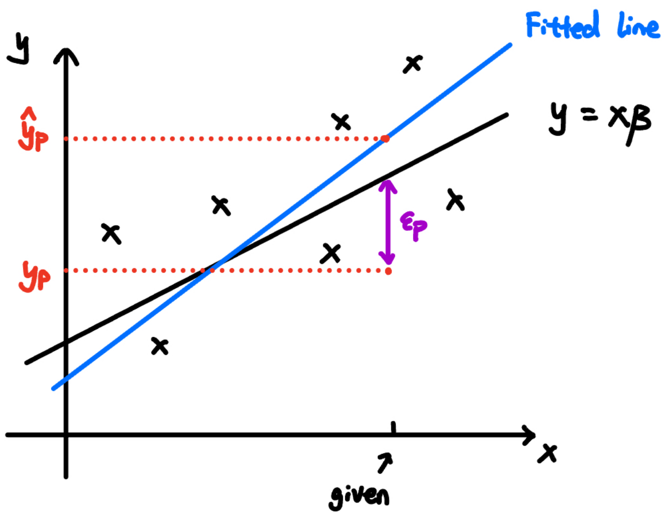
$$\begin{aligned}
\operatorname{Var}\left(y_p - \hat{y}_p\right) &= \operatorname{Var}\left(\mathbf{a}_p^\top \boldsymbol{\beta} + \epsilon_p - \mathbf{a}_p^\top \hat{\boldsymbol{\beta}}\right) \\
&= \operatorname{Var}\left(\epsilon_p - \mathbf{a}_p^\top \hat{\boldsymbol{\beta}}\right) && \text{since } \mathbf{a}_p^\top \boldsymbol{\beta} \text{ is a constant} \\
&= \operatorname{Var}\left(\epsilon_p\right) + \operatorname{Var}\left(\mathbf{a}_p^\top \hat{\boldsymbol{\beta}}\right) && \text{since } \epsilon_p \perp\!\!\!\perp \hat{\boldsymbol{\beta}} \\
&= \sigma^2 + \sigma^2 \mathbf{a}_p^\top (X^\top X)^{-1} \mathbf{a}_p \\
&= \sigma^2 \left[1 + \mathbf{a}_p^\top (X^\top X)^{-1} \mathbf{a}_p\right].
\end{aligned}$$

As usual, we plug in $\hat{\sigma}^2 = \frac{\text{SSE}}{n-p-1}$ in place of σ^2. Thus,

$$\frac{y_p - \hat{y}_p}{\hat{\sigma}\sqrt{1 + \mathbf{a}_p^\top (X^\top X)^{-1} \mathbf{a}_p}} \sim t_{n-p-1}.$$

Proposition 2.9 (Prediction Interval for y_p).
A $100(1-\alpha)\%$ prediction interval for y_p is

$$\hat{y}_p \pm t_{n-p-1,\alpha/2}\,\text{SE}\left(\hat{y}_p - y_p\right) \quad \text{where } \text{SE}\left(\hat{y}_p - y_p\right) = \hat{\sigma}\sqrt{1 + \mathbf{a}_p^\top (X^\top X)^{-1} \mathbf{a}_p}.$$



MLR in R

```
dat <- read.csv("lowbwt2.csv", header=T)
dim(dat)
head(dat)

pairs(dat)

library(car)
car::scatterplotMatrix(dat, col="black")

## Matrix form
X <- dat[,2:4]  # Y = βX + epsilon
dim(X)

X <- as.matrix(cbind(1, X))   # appends a column of 1's in the front
dim(X)
head(X)

n <- nrow(X)
p <- ncol(X) - 1
n
p

Y <- dat$headcirc   # equivalently, dat[,1]
Ybar <- mean(Y)

## LSE
beta <- solve(t(X) %*% X) %*% t(X) %*% Y  # β = (X^t X)^{-1} X^t Y
beta

## Hat matrix
```

```
H <- X %*% solve(t(X) %*% X) %*% t(X)    # H = X(X^t X)^{-1} X^t
dim(H)
# Idempotent
max(abs(H %*% H - H))    # HH - H ≈ 0, very close to 0
# symmetric
max(abs(t(H) - H))  # H^t - H ≈ 0
Yhat <- H %*% Y    # Yhat = HY

## Estimate σ^2
resid <- Y-Yhat
SSE <- sum(resid^2)
sigma2 <- SSE/(n-p-1)
sigma <- sqrt(sigma2)
c(sigma, sigma2)

## Estimate variance-covariance matrix of βhat
XtX <- solve(t(X) %*% X)
XtX
V <- sigma2 * XtX
V

## Confidence interval for β1
beta[2] + c(-1, 1)*qt(0.975, n-p-1)* sqrt(V[2,2])

## Prediction interval
#    for a child with gestage=30, birthwt=1200, toxemia=0
a <- c(1,30,1200,0)
a <- as.matrix(a, nrow=4, ncol=1)
# estimate
t(a) %*% beta

# variance
t(a) %*% V %*% a
```

```r
# final prediction interval
c(t(a) %*% beta) + c(-1, 1)*qt(0.975, n-p-1)*
c(sqrt(1+t(a) %*% XtX %*% a))*sigma

### MLR using lm()
fm <- lm(headcirc~gestage+birthwt+toxemia, data=dat)
smy <- summary(fm)
names(smy)
coef(smy)

## Confidence interval for β1
smy$coefficients
smy$coefficients[2,2]
coef(fm)[2] + c(-1, 1)*qt(0.975, n-p-1)*smy$coefficients[2,2]
# Alternatively: direct method
confint(fm, 2)

## Estimate variance-covariance matrix of βhat
smy$cov.unscaled
smy$^sigma2 * smy$cov.unscaled

## Prediction interval
#     for a child with gestage=30, birthwt=1200, toxemia=0
newdata <- data.frame(gestage=30, birthwt=1200, toxemia=0)
predict(fm, newdata)
predict(fm, newdata, interval="prediction", level=0.95)
```

Lecture 11, 2025/10/09

2.5 MLR: ANOVA

MLR model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \ldots, n.$$

The sum of squares of residuals (error) is

$$\mathrm{SSE}(\hat{\boldsymbol{\beta}}) = \sum_{i=1}^{n} r_i^2 = \mathbf{r}^\top \mathbf{r} = (\mathbf{Y} - X\hat{\boldsymbol{\beta}})^\top (\mathbf{Y} - X\hat{\boldsymbol{\beta}}).$$

Test $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$, recall that

$$\frac{\hat{\beta}_i - \beta_i}{\mathrm{SE}\left(\hat{\beta}_i\right)} \sim t_{n-p-1}.$$

Question: What if we reject H_0 if we reject any $H_{0i} : \beta_i = 0$ for $i = 1, \ldots, p$ at level $\alpha = 0.05$?

Assume H_0 is true and $\hat{\beta}_i$'s are independent. Then,

$$\mathbb{P}\left(\text{reject } H_0 \mid H_0 \text{ is true}\right) = \mathbb{P}\left(\text{reject any } H_{0i} \mid H_0\right) = 1 - \mathbb{P}\left(\text{accept all } H_{0i} \mid H_0\right) = 1 - (1 - 0.05)^p.$$

Under H_0, the full model is reduced to:

$$y_i = \beta_0 + \epsilon_i, \quad i = 1, \ldots, n.$$

This is called the **reduced model** or **restricted model**. The LSE under the reduced model is

$$\hat{\beta}_0 = \bar{y},$$

and

$$\mathrm{SSE}(\hat{\beta}_0) = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \mathrm{SST}\,.$$

Consider the following difference:

$$\mathrm{SSE}(\hat{\beta}_0) - \mathrm{SSE}(\hat{\boldsymbol{\beta}}) = \mathrm{SST} - \mathrm{SSE}(\hat{\boldsymbol{\beta}}) = \mathrm{SSR}$$

where

$$
\begin{aligned}
\mathrm{SSR} &= \sum_{i=1}^{n}(y_i-\bar{y})^2-(\mathbf{Y}-X\hat{\boldsymbol{\beta}})^\top(\mathbf{Y}-X\hat{\boldsymbol{\beta}})\\
&= \sum_{i=1}^{n} y_i^2-n\bar{y}^2-\left[(I-H)\mathbf{Y}\right]^\top\left[(I-H)\mathbf{Y}\right]\\
&= \mathbf{Y}^\top\mathbf{Y}-n\bar{y}^2-\mathbf{Y}^\top(I-H)^\top(I-H)\mathbf{Y}\\
&= \mathbf{Y}^\top\mathbf{Y}-n\bar{y}^2-\mathbf{Y}^\top(I-H)\mathbf{Y} && \text{since } I-H \text{ is symmetric and idempotent}\\
&= \mathbf{Y}^\top\mathbf{Y}-n\bar{y}^2-\mathbf{Y}^\top\mathbf{Y}+\mathbf{Y}^\top H\mathbf{Y}\\
&= \mathbf{Y}^\top H\mathbf{Y}-n\bar{y}^2\\
&= \mathbf{Y}^\top H^\top H\mathbf{Y}-n\bar{y}^2 && \text{since } H \text{ is symmetric and idempotent}\\
&= \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}}-n\bar{y}^2\\
&= \hat{\boldsymbol{\beta}}^\top X^\top X\hat{\boldsymbol{\beta}}-n\bar{y}^2.
\end{aligned}
$$

Proposition 2.10. Under $H_0 : \beta_1 = \cdots = \beta_p = 0$, we have

$$\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_p.$$

Using this result, we can construct the F-statistic:

$$F = \frac{\mathrm{SSR}\,/p}{\mathrm{SSE}\,/(n-p-1)} = \frac{\mathrm{MSR}}{\mathrm{MSE}} \sim F_{p,n-p-1}.$$

We reject H_0 at level α if

$$F > F_\alpha(p, n-p-1), \quad \text{where } \alpha \text{ is the } 100(1-\alpha)\%\text{-tile.}$$

We have the following ANOVA Table:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F
Regression	$\text{SSR} = \hat{\boldsymbol{\beta}}^\top X^\top X \hat{\boldsymbol{\beta}} - n\bar{y}^2$	p	$\text{MSR} = \dfrac{\text{SSR}}{p}$	$\frac{\text{MSR}}{\text{MSE}}$
Residual	$\text{SSE} = (\mathbf{Y} - X\hat{\boldsymbol{\beta}})^\top (\mathbf{Y} - X\hat{\boldsymbol{\beta}})$	$n - p - 1$	$\text{MSE} = \dfrac{\text{SSE}}{n-p-1}$	
Total	$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})$	$n - 1$		

Definition 2.12 (Total Coefficient of Determination).

THe **total coefficient of determination** is

$$R^2 = \frac{\text{SSR}}{\text{SST}}$$

where $0 \leq R^2 \leq 1$.

Recall the MLR model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip} + \epsilon_i$$

and $y_i \sim \mathrm{N}\left(\beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip}, \sigma^2\right)$ independently. The PDF of $y \sim \mathrm{N}(\mu, \sigma^2)$ is

$$f(y) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right).$$

The likelihood function is

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n f(y_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{\left[y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right]^2}{2\sigma^2}\right).$$

The log-likelihood function is

$$\begin{aligned}
\ell(\boldsymbol{\beta}, \sigma^2) &= \log L(\boldsymbol{\beta}, \sigma^2) \\
&= \sum_{i=1}^{n} \left(\log\left(2\pi\sigma^2\right)^{-1/2} - \frac{\left[y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right]^2}{2\sigma^2} \right) \\
&= -\frac{n}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^{n} \left[y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right]^2}_{S(\boldsymbol{\beta})}.
\end{aligned}$$

For $\boldsymbol{\beta}$, maximizing $\ell(\boldsymbol{\beta}, \sigma^2) \iff$ minimizing $S(\boldsymbol{\beta})$. That is,

$$\hat{\boldsymbol{\beta}}_{\text{MLE}} = \hat{\boldsymbol{\beta}}_{\text{LSE}}, \quad \text{and} \quad S(\hat{\boldsymbol{\beta}}) = \text{SSE}.$$

For σ^2,

$$\frac{\partial \ell(\boldsymbol{\beta}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2} \cdot \frac{1}{2\pi\sigma^2} 2\pi + \frac{\text{SSE}}{2} \cdot \frac{1}{(\sigma^2)^2} = 0 \implies -n + \frac{\text{SSE}}{\sigma^2} = 0 \implies \hat{\sigma}^2_{\text{MLE}} = \frac{\text{SSE}}{n}.$$

Lecture 12, 2025/10/21

2.6 Geometric Interpretation of LSE

Recall that $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$ represents n simultaneous equations

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n.$$

Note that $X = \begin{bmatrix} \vec{1} & \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_p \end{bmatrix}$ where

$$\vec{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{bmatrix} \text{ for } i = 1, \dots, p.$$

The systematic part

$$
\begin{aligned}
X\boldsymbol{\beta} &= \begin{bmatrix} \vec{1} & \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_p \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \\
&= \beta_0 \vec{1} + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \cdots + \beta_p \mathbf{x}_p \\
&= \sum_{i=0}^{p} \beta_i \mathbf{x}_i \quad \text{where } \mathbf{x}_0 = \vec{1}
\end{aligned}
$$

is a linear combination of the vectors $\mathbf{x}_i$.

Definition 2.13 (Column Space).

The **column space** of X is defined as

$$
\begin{aligned}
C(X) &= \{\text{all vectors that can be constructed as a linear combination of the columns of } X\} \\
&= \left\{ X\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^{p+1} \right\} \\
&= \left\{ \beta_0 \vec{1} + \beta_1 \mathbf{x}_1 + \cdots + \beta_p \mathbf{x}_p : \beta_i \in \mathbb{R}, i = 0, 1, \ldots, p \right\}.
\end{aligned}
$$

Note. $\hat{\mathbf{Y}} = X\hat{\boldsymbol{\beta}} \in C(X)$.

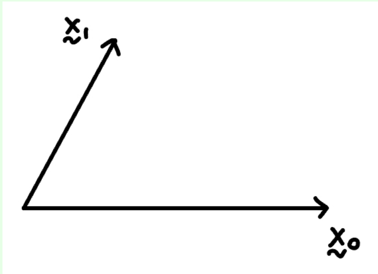
$C(X)$ spans a $p+1$-dimensional subspace inside the n-dimensional space (e.g. $\mathbb{R}^n$).

Note that

$$
\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}
$$

is a point in a n-dimensional space and represents a vector. The same is true for $\vec{1}, \mathbf{x}_1, \ldots, \mathbf{x}_p$.

Example. $\beta_0 \mathbf{x}_0 + \beta_1 \mathbf{x}_1$ is a linear combination of two vectors $\mathbf{x}_0$ and $\mathbf{x}_1$, spanning a plane.



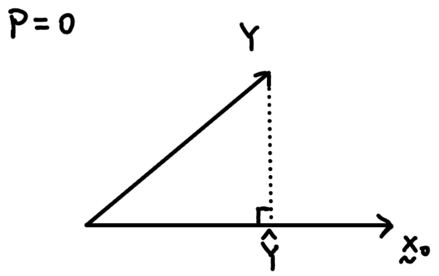
The LSE $\hat{\beta}$ minimizes

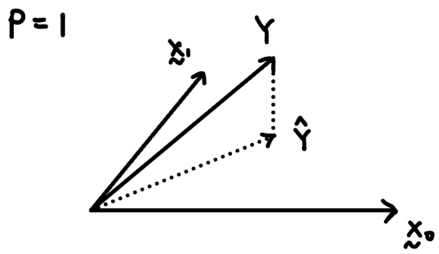
$$\text{SSE} = \sum_{i=1}^{n} r_i^2,$$

the length $\sqrt{r_1^2 + r_2^2 + \cdots + r_n^2}$ of the residual vector

$$\mathbf{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}$$

subject to the constraint $\hat{\mathbf{Y}} \in C(X)$. Recall that $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{r}$. The LSE tries to find $\hat{\mathbf{Y}} \in C(X)$ that is "closest" to $\mathbf{Y}$.





Note that $\hat{\mathbf{Y}} = H\mathbf{Y}$ is the **perpendicular projection** of $\mathbf{Y}$ onto $C(X)$.

Since $\mathbf{r} \perp C(X)$, or equivalently,

$$\begin{aligned}
\vec{1}^\top \mathbf{r} &= 0, \\
\mathbf{x}_1^\top \mathbf{r} &= 0, \\
&\vdots \\
\mathbf{x}_p^\top \mathbf{r} &= 0.
\end{aligned}$$

(inner product $= 0$). This implies that

$$\begin{aligned}
X^\top \mathbf{r} &= \vec{0} \\
X^\top (\mathbf{Y} - X\hat{\boldsymbol{\beta}}) &= \vec{0} \\
X^\top X\hat{\boldsymbol{\beta}} &= X^\top \mathbf{Y} \\
\hat{\boldsymbol{\beta}} &= (X^\top X)^{-1} X^\top \mathbf{Y}.
\end{aligned}$$

Proposition 2.11.

(1) $\mathbf{Y}$ is projected onto two orthogonal subspaces $C(X)$ and $C(X)^\perp$:

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{r}, \quad \hat{\mathbf{Y}} \in C(X), \quad \mathbf{r} \in C(X)^\perp$$

where $C(X)$ has dimension $p+1$ and $C(X)^\perp$ has dimension $n-p-1$.

(2) Under the normality assumption, $\hat{\mathbf{Y}}$ and $\mathbf{r}$ are independent, i.e. $\hat{\mathbf{Y}} \perp\!\!\!\perp \mathbf{r}$.

Test Linear Constraints

Example. Test $H_0 : \beta_q = \beta_{q+1} = \cdots = \beta_p = 0$ where $1 \le q \le p$. We can write H_0 in the form of

$$A\boldsymbol{\beta} = \vec{0}$$

where

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}_{(p-q+1)\times(p+1)} .$$

The first 1 entry in the first row is in the $(q+1)$th column.

Example. Consider

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

and the literature suggests

$$\beta_1 = 2\beta_2, \quad \text{and} \quad \beta_3 = 0.$$

We can write this in matrix form $A\boldsymbol{\beta} = \vec{0}$ where

$$A = \begin{bmatrix} 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Suppose A is a $\ell \times (p+1)$ matrix, i.e. we have ℓ linear constraints. Our test procedure relies on the **additional sum of squares principle**. Re call that

$$C(X) = \{\beta_0 \vec{1} + \beta_1 \mathbf{x}_1 + \cdots + \beta_p \mathbf{x}_p\}$$

We define

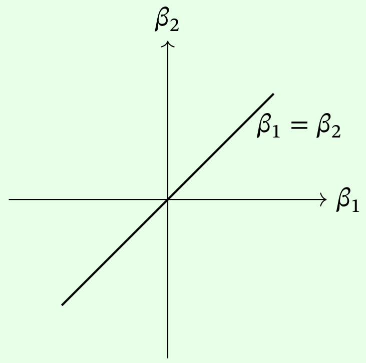
$$C_A(X) = \{\beta_0 \vec{1} + \beta_1 \mathbf{x}_1 + \cdots + \beta_p \mathbf{x}_p \,:\, A\boldsymbol{\beta} = \vec{0}\} \subseteq C(X)$$

which is a subspace of $C(X)$.

Example. For

$$\beta_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}.$$

The linear constraint is $\beta_1 = \beta_2$.



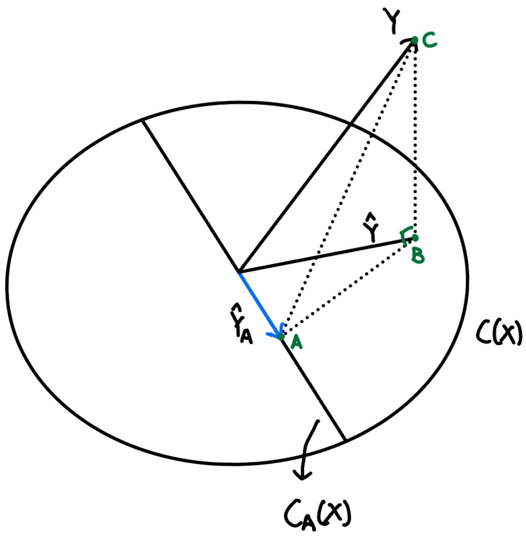
Let $\hat{\mathbf{Y}}$ be the orthogonal projection of Y onto $C(X)$, and let $\hat{\mathbf{Y}}_A$ be the orthogonal projection of Y onto $C_A(X)$. If $H_0 : A\boldsymbol{\beta} = \vec{0}$ is true, we expect $\hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}}_A$ to be similar/close. That is, the model makes similar predictions whether we set $A\boldsymbol{\beta} = \vec{0}$ or not when fitting the model.

The squared distance between $\hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}}_A$

$$\left\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A\right\|^2 = (\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A)^\top(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A)$$

is the **additional sum of squares**.

Geometric representation of $C(X)$ and $C_A(X)$:



By Pythagoras' theorem,

$$(AC)^2 = \text{SSE}_A, \quad (BC)^2 = \text{SSE}$$

The squared distance is

$$\left\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A\right\|^2 = (AB)^2 = (AC)^2 - (BC)^2 = \text{SSE}_A - \text{SSE}$$

where $\text{SSE}_A = \left\|\mathbf{Y} - \hat{\mathbf{Y}}_A\right\|^2$ and $\text{SSE} = \left\|\mathbf{Y} - \hat{\mathbf{Y}}\right\|^2$.

Theorem 2.12. Under $H_0 : A\boldsymbol{\beta} = \vec{0}$ where A is a $\ell \times (p+1)$ matrix, we have

(1) $\frac{\left\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A\right\|^2}{\sigma^2} \sim \chi^2_\ell$.

(2) $\left\|\mathbf{Y} - \hat{\mathbf{Y}}\right\|^2$ is independent of

$$\hat{\sigma}^2 = \frac{(\mathbf{Y} - \hat{\mathbf{Y}})^\top(\mathbf{Y} - \hat{\mathbf{Y}})}{n-p-1}.$$

That is, $\left\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A\right\|^2 \perp\!\!\!\perp \hat{\sigma}^2$.

Lecture 13, 2025/10/23

The theorem above implies the following F-statistic (under $H_0 : A\boldsymbol{\beta} = \vec{0}$):

$$F = \frac{\left\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_A\right\|^2/\ell}{\hat{\sigma}^2} \sim F(\ell, n-p-1).$$

Note. We saw earlier that the ratio of two independent chi-square random variables leads to an F-distribution. Large values of F provide evidence against H_0.

We reject H_0 at level α if

$$F > F_\alpha(\ell, n-p-1),$$

or we compute the p-value as

$$p\text{-value} = \mathbb{P}\left(F(\ell, n-p-1) > F\right).$$

In summary, to test $H_0 : A\boldsymbol{\beta} = \vec{0}$, we follow these steps:

(1) Fit the full model without constraints.

(2) Compute SSE.

(3) Fit the restricted model with constraints $A\boldsymbol{\beta} = \vec{0}$.

(4) Compute SSE_A, the sum of squares residuals under the restricted model.

(5) Compute the F-statistic:

$$F = \frac{(\text{SSE}_A - \text{SSE})/\ell}{\text{SSE}/(n-p-1)}.$$

Question: How to fit the restricted model with constraints $A\boldsymbol{\beta} = \vec{0}$?

Example (Continued).

Back to Example 2.6, we have

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

with $\beta_1 = 2\beta_2$ and $\beta_3 = 0$. Thus,

$$y = \beta_0 + 2\beta_2 x_1 + \beta_2 x_2 + \epsilon = \beta_0 + \beta_2(2x_1 + x_2) + \epsilon.$$

Case Study: UFFI Example